

MTH203: Sequences and series

Tutorial/Homework 03

15/09/2025

Due date: 22/09/2025, 6:00 PM

Problem 1. For any real number x , let $[x]$ denote the greatest integer less than or equal to x . Compute $\lim_{n \rightarrow \infty} [5/n]$.

Problem 2. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. Prove that

$$\lim_{n \rightarrow \infty} |x_n| = \left| \lim_{n \rightarrow \infty} x_n \right|.$$

Problem 3. If $\lim_{n \rightarrow \infty} a_n = 0$ and $|b_n - b| \leq a_n$ for all $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} b_n = b$.

Problem 4. For every *non-negative* real number t , let $\sqrt{t}$ denote the unique non-negative real number s such that $s^2 = t$.

- (a) Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of non-negative real numbers such that $\lim_{n \rightarrow \infty} x_n = 0$. Prove that $\lim_{n \rightarrow \infty} \sqrt{x_n} = 0$.
- (b) Let x be a *positive* real number. Prove that if $(x_n)_{n \in \mathbb{N}}$ is a sequence of non-negative real numbers such that $\lim_{n \rightarrow \infty} x_n = x$, then $\lim_{n \rightarrow \infty} \sqrt{x_n} = \sqrt{x}$. (Hint: You could use the fact that $\sqrt{x} - \sqrt{x_n} = \frac{x - x_n}{\sqrt{x} + \sqrt{x_n}}$.)